[I salute the young students for their peaceful and exemplary behaviour in our capital inspite of the brutality unleashed by our rulers — you were branded terrorists and viruses; you were forced to sacrifice precious five weeks of your life; you had to bear the unethical snooping and surveillance by our savage rulers; you had to resignedly accept the police-seized damaged van, loaded with stones, parked near you with the sole intention of provoking and implicating you; you suffered taser shocks, nail studded sticks, tear gas, pellet guns of our barbaric rulers (till the hospitals in Kashmir stopped giving data, these pellets blinded more than a thousand in the past). You maintained civility all through, hats off to you. While all this by our dishonourable rulers was going on, honourable gentlemen entrusted by us to deliver justice, shamelessly said 'do not waste our time'. With an evaporated justice system in the country, You now face trolling, intimidation, FIRs by our thuggish rulers and their slaves. I feel your pain.]

Just to bring you to the working mood, some useful problems; but irrelevant for our course.

1. Show that there are no real numbers a and b such that $\frac{1}{a} + \frac{1}{b} = \frac{1}{a+b}$ But there are complex numbers satisfying it.

2. Let C be the set of all pairs (a, b) of real numbers with usual addition $(+)$ and multiplication given by $(a, b) \times (c, d) = (ac - bd, ad + bc)$. Show that $(C, +, \times)$ is a field.

   Let M be the set of all 2×2 matrices of the form $\begin{pmatrix} a & b \\ -b & a \end{pmatrix}$ where a, b are real numbers. With usual addition and multiplication of matrices, show that it forms a field.

   Let Q be the set of all polynomials $p(t)$ in one variable t with real coefficients. Consider usual addition and multiplication of polynomials. Say $p_1 \sim p_2$ if there is a polynomial p such that $p_1 - p_2 = (t^2 + 1)p$. Let P be the set of equivalence classes. Show that the equivalence relation respects (?) the operations of addition and multiplication. Show that P is a field.

   Show that the three fields above are isomorphic in a 'canonical way'.

   For even integer $n \geq 1$, show that there are $n \times n$ invertible real matrices A and B such that

$$A^{-1} + B^{-1} = (A + B)^{-1}.$$

3. Let Q be R^4 with points written as $(a,b,c,d) = a + bi + cj + dk$. Use coordinate-wise addition. Multiplication is 'prescribed' by
$i^2 = j^2 = k^2 = ijk = -1$.
Show $ij = k, jk = i, ki = j$ and $ji = -k, kj = -i, ik = -j$.
Show that Q is nearly a field, it misses only commutativity of multiplication. It is called skew-field.
Show M, the set of 4×4 real matrices of the form $\begin{pmatrix} a & -b & -c & -d \\ b & a & d & -c \\ c & -d & a & b \\ d & c & -b & a \end{pmatrix}$ is 'canonically' isomorphic to the above skew field. This skew field is called 'Quaternions.'

4. Let a and b be real numbers. Show

$$\int_0^\infty \frac{1}{1+x^2}\,\frac{x^b - x^a}{(1+x^a)(1+x^b)}\,dx \;=\; 0.$$

Deduce

$$\int_0^\infty \frac{1}{1+x^2}\,\frac{1}{1+x^a}\,dx \;= \frac{1}{2}\int_0^\infty \frac{1}{1+x^2}\,dx \;=\; \frac{\pi}{4}.$$

5. If $I_n = (x^2+1)^{-n}dx$, show

$$I_n = \frac{x}{2(n-1)(x^2+1)^{n-1}} + \frac{2n-3}{2(n-1)} I_{n-1}.$$

6. Show

$$\frac{1}{x^4+1} = \frac{1}{2\sqrt{2}}\,\frac{x+\sqrt{2}}{x^2+\sqrt{2}\,x+1} \;-\; \frac{1}{2\sqrt{2}}\,\frac{x-\sqrt{2}}{x^2-\sqrt{2}\,x+1}$$

Hence argue

$$\int \frac{dx}{x^4+1} = \frac{1}{4\sqrt{2}}\,\log|x^2+\sqrt{2}\,x+1| \;-\; \frac{1}{4\sqrt{2}}\,\log|x^2-\sqrt{2}\,x+1|$$

$$+\,\frac{1}{2\sqrt{2}}\,\tan^{-1}(\sqrt{2}\,x+1) \;+\; \frac{1}{2\sqrt{2}}\,\tan^{-1}(\sqrt{2}\,x-1)$$

7. Fibonacci numbers are defined by $F_0 = F_1 = 1; F_{n+2} = F_{n+1} + F_n$ for $n \geq 0$. Show $\sum_0^\infty F_n t^n \;=\; \frac{1}{1-t-t^2}$.

8. Let f and g be functions on R to R which are differentiable as many times as needed below. Show Leibniz's rule:

$$D^p(fg) \;=\; \sum_{k=0}^{p} \binom{p}{k} D^k f \; D^{p-k} g.$$

Calculate the first ten Bernoulli numbers defined by

$$\frac{t}{e^t - 1} \;=\; \sum_{0}^{\infty} \frac{B_n}{n!} t^n.$$

Show

$$\sum_{j=0}^{n-1} e^{jt} \;=\; \frac{e^{nt} - 1}{t} \; \frac{t}{e^t - 1} \;=\; f(t)g(t) \quad \text{say.}$$

Show

$$\sum_{j=0}^{n-1} j^p \;=\; \sum_{k=0}^{p} \binom{p}{k} (D^k f)(0) \; (D^{p-k} g)(0).$$

$$= \sum_{k=0}^{p} \binom{p}{k} \frac{n^{k+1}}{k+1} \; B_{p-k} \;=\; \sum_{k=0}^{p} \binom{p+1}{k+1} \frac{n^{k+1}}{p+1} \; B_{p-k}.$$

Deduce

$$1 + 2 + 3 + \cdots + n \;=\; \frac{1}{2}n^2 + \frac{1}{2}n.$$

$$1^2 + 2^2 + 3^2 + \cdots + n^2 \;=\; \frac{1}{3}n^3 + \frac{1}{2}n^2 + \frac{1}{6}n.$$

$$1^3 + 2^3 + 3^3 + \cdots + n^3 \;=\; \frac{1}{4}n^4 + \frac{1}{2}n^3 + \frac{1}{4}n^2.$$

$$1^{10} + 2^{10} + 3^{10} + \cdots + n^{10} \;=$$

$$\frac{1}{11}n^{11} + \frac{1}{2}n^{10} + \frac{5}{6}n^9 - n^7 + n^5 - \frac{1}{2}n^3 + \frac{5}{66}n.$$

[who is a bigger cheat than a person who uses bad words against peaceful students and then inflicts lasting wounds using barbaric methods – finally says he has forgiven them. What is a bigger fake speech than asking students (in a convocation) to question things while people that question are put in Jail without trial. It is frustrating to me when CJI says students and youth should be advised and counselled. Do they think peaceful protest is wrong. Why did the CJI not utter one, one, word about using pellets and other methods on peaceful students; instead, said 'do not waste our time'. Why did he not utter one word on the harassment of decent youth by our super-bigoted rulers and their slaves. As a nationalist, I can only pray that honesty and civility and fairness and respect for truth – which dried up in our halls of justice – return soon. I am not holding courts in contempt, I am saying: some judges are doing it amply.]

most problems below are routine, a few do need work.

9. Let X be any set and $P(X)$ be its power set, that is, collection of all subsets of X. Define for $A, B \subset X$, their symmetric difference, by.
$A\Delta B = (A - B) \cup (B - A) = \{x : x \quad \text{is in exactly one of the sets.}\}$.
Show that $A_1\Delta A_2\Delta A_3\Delta \cdots \Delta A_n$ consists of points that belong to an odd number of these sets; in whichever order you operate.
Show that $P(X)$ is a group under the operation Δ. Let $H \subset P(X)$ be the collection of finite sets. Show that H is a subgroup. Let $H \subset P(X)$ be the collection of countable sets. Show that H is a subgroup. If H is the collection of countably infinite sets then is it a subgroup.

10. Let G be a group and X a non-empty set. The collection of functions on X to G, denoted by G^X, is a group with pointwise (?) operations.

    The set $2 = \{0, 1\}$ is a group under addition modulo 2. What is the relation between the group 2^X and the earlier $P(X)$ group.

11. You can define union of any family of sets. Suppose $\mathcal{C}$ is a collection of sets. Then their union, $\cup\mathcal{C}$, is the set of all objects x such that $x \in C$ for some sets $C \in \mathcal{C}$. Similarly, $\cap\mathcal{C}$ is the collection of all objects x such that $x \in C$ for every $C \in \mathcal{C}$. Prove DeMorgan's laws:$(\cup\mathcal{C})^c = \cap\mathcal{C}^c$; $(\cap\mathcal{C})^c = \cup\mathcal{C}^c$ where $\mathcal{C}^c = \{C^c : C \in \mathcal{C}\}$.

12. For a sequence of sets (A_n), we define $\limsup A_n$ to be the set of all objects x such that x belongs to A_n for infinitely many values of n.

When this happens, we also say that $x \in A_n$ frequently. Similarly, $\liminf A_n$ is the set of all objects x $x \in A_n$ for every $n \geq n_0$ for some n_0. When this happens we also say that $x \in A_n$ eventually. Prove De Morgan's laws: $(\limsup A_n)^c = \liminf A_n^c$; $(\liminf A_n)^c = \limsup A_n^c$..

If X is the universe of discourse (this means all sets we now consider in this exercise are subsets of this X), we define for a set $A \subset X$ its indicator function or characteristic function to be the following function defined on X: $I_A(x) = 1$ if $x \in A$ and $I_A(x) = 0$ if $x \notin A$. This function I_A is also denoted by χ_A.

Show that $\limsup I_{A_n} = I_{\limsup A_n}$; $\liminf I_{A_n} = I_{\liminf A_n}$

Let $A_n = [0, 1]$ if n is even and $A_n = [2, 3]$ if n is odd. Find limsup and liminf.

Let $\{a_n\}$ be a sequence of real numbers and $A_n = (-\infty, a_n)$ and $B_n = (-\infty, a_n]$. Calculate $\liminf A_n$ and $\limsup A_n$ in terms of liminf and limsup of a_n. Do the same for (B_n).

13. Write a complete proof of the fact: If $m < n$ then there is no bijection between $\{1, 2, \cdots, m\}$ and $\{1, 2, , \cdots, n\}$.

    If X is a countable set, then show that there is a $1-1$ function on X to (into or onto) N. Conversely, if there is a $1-1$ function on X to N then show that X is countable.

    If X is countable set, show that there is a function on N onto X. Conversely, if there is a function on N onto X, show that X is countable.

    Let X be a countable set and $Y \subset X$. Show that Y is countable.

    Let X_i for $i = 1, 2, \cdots$ be countable sets. Show that $\cup_i X_i$ is a countable set. This is stated as 'countable union of countable sets is countable'.

    If X and Y are countable, show that $X \times Y$ is also countable.

    Show seq(N), defined as the set of all finite sequences of integers, is countable. Let Y be the set of all infinite sequences of integers. Show Y is not countable.

14. If X is the set of all 5×5 real matrices and $Y = R^{25}$, exhibit a canonical bijection and conclude $|X| = |Y|$. If X is the set of all real symmetric 5×5 matrices and $Y = R^{15}$ show that $|X| = |Y|$. If Z is the set of all upper triangular 5×5 matrices then $|Z| = |X|$.

15. If X is an infinite set, prove there is a subset $Y \subset X$ which is countably infinite. If X is an infinite set then show that there is a proper subset $Y \subset X$ and $Y \neq X$ such that $|X| = |Y|$. Conversely, if X is a set and if there is a proper subset $Y \subset X$ and $Y \neq X$ such that $|X| = |Y|$ then show that X is an infinite set.

16. Let Y be the set of all ordered pairs of real numbers, in other words, $Y = R^2$. Show $|Y| = |R|$. More generally show that $|R^n| = |R|$ for any integer $n \geq 1$. Let seq(R) be set of all finite sequences of real numbers and $X =$ the set of all infinite sequences of real numbers. Show both have same cardinality as R.

17. Verify if the following are equivalence relations on the sets prescribed.
$X = R$. $a \sim b$ if $|a - b| \leq 25$.
$X = R$. $a \sim b$ if $a - b$ is an integer. Or $a \sim b$ if $a - b$ is rational. Or $a \sim b$ if $a - b$ is irrational.
Let $X = P(R)$. Say $A \sim B$ if $A \Delta B$ is finite. What if finite is replaced by countable? What if finite is replaced by countably infinite?
Let X be the set of all functions from R to R. Say $f \sim g$ if $f - g$ is a continuous function. What if 'continuous function' is replaced by 'polynomial'.
Let $X = R^{29}$. Say $a = (a_1, \cdots, a_{29}) \sim b = (b_1, \cdots, b_{29})$ if there is a permutation π of $\{1, 2, \cdots, 29\}$ such that $a_i = b_{\pi(i)}$ for all i.
Same X as above. Say $a \sim b$ if there is a permutation π of $\{1, 2, \cdots, 25\}$ such that $a_i = b_{\pi(i)}$ for all i with $1 \leq i \leq 25$.
Same X as above, say $a \equiv b$ if $\sum a_i = \sum b_i$.

18. (optional challenge, not part of course)

$N = \{1, 2, \cdots\}$. $\mathcal{U} \subset P(N)$ is called an ultrafilter if it satisfies the following conditions: (i) $N \in \mathcal{U}$; $\quad \emptyset \notin \mathcal{U}$; (ii) $A, B \in \mathcal{U} \Rightarrow A \cap B \in \mathcal{U}$; (iii) $A \in \mathcal{U}, A \subset B \Rightarrow B \in \mathcal{U}$; (iv) $A \subset N \Rightarrow A \in \mathcal{U}$ or $A^c \in \mathcal{U}$.
Thus $\mathcal{U}$ is closed under finite intersections, supersets, has N and not the empty set and for all sets either that or its complement belongs.
A collection $\mathcal{F}$ satisfying only the first three conditions is called a filter. Here are examples of filters. Verify. (i) All sets A such that A^c is finite. This is called co-finite filter. (ii) All sets A such that A^c has only finitely many even numbers. (iii) All supersets of $\{23, 48, 56\}$ Verify that None of these is ultrafilter.
Verify: All sets that contain the number 103 is ultrafilter. You (and me too) can not think of other ultrafilters! — except replacing 103 by another number. These ultrafilters as above are called fixed ultrafilters. Use Zorn to verify there are ultrafilters which are not fixed, they are called free ultrafilters. Show that an ultrafilter is free iff it misses all singleton sets. Show that an ultrafilter is free iff it includes co-finite filter.

Success is not final, failure is not fatal: It is the courage to continue that counts. Winston Churchill

I've failed over and over and over again in my life. And that is why I succeed. To learn to succeed, you must first learn to fail. I can accept failure, everyone fails at something. But I can't accept not trying.

Michael Jordan

Creativity is present in each and everyone of us due to us being humans and having the ability to think and innovate. It is basic human nature to be curious, to try new things, and to learn by discovering new connections.Creative thinking involves leaving the known and entering the unknown. It is about burning new neural connections in the brain so that creative ideas have the opportunity to surface. Aditya Shanbhag

19. Let $U \subset R$ be open. Show that $|U|$ is either 0 or c. Let X be the space of all infinite sequences $(x_1, x_2, \cdots)$ where $0 < x_i < 1$ for each i . Show $|X| = c$.

20. Show $[0,1] \setminus C = (0,1) \setminus C$ is a disjoint union of open intervals and let $\mathcal{I}$ is the collection of exactly these open intervals. equivalently $\mathcal{I}$ is the collection of all deleted (while making C) open intervals of $(0,1)$.

    If $I = (a,b)$ and $J = (c,d)$ are in $\mathcal{I}$ then say that $I \leq J$ if either they are the same interval or $b < c$ (that is, the interval I sits to the left of J). Show that the collection $\mathcal{I}$ with this order is another manifestation (?) of Q.

21. How large a collection of non-empty subsets of N can you get so that any two are disjoint? That is, I want a family of sets $(A_\alpha : \alpha \in \Delta)$, each $\emptyset \neq A_\alpha \subset N$; such that for distinct indices $A_\alpha \cap A_\beta = \emptyset$. How large can the set Δ be? Answer: $|\Delta| = \aleph_0$.
    What if we need the non-empty sets to be only nearly disjoint? That is, I want a family $(A_\alpha; \alpha \in \Delta)$ of non-empty subsets of N so that for distinct indices $A_\alpha \cap A_\beta$ is a *finite set*, could be but need not be empty. How large can Δ be? Answer: $|\Delta| = c$. To see this, First argue $|\Delta| \leq c$. Fix an enumeration $\{r_1, r_2, \cdots\}$ of rational numbers. You can do this explicitly without using AC (but this is beside the point). For $x \in R$, let us define a set as follows. x_1 is the first rational in $(x-1, x+1)$. x_2 is the first rational that occurs after x_1 which is in the interval $(x - 1/2, x + 1/2)$. In general x_k is the first rational that occurs after

x_{k-1} which is in the interval $(x-1/k, x+1/k)$. Let $A_x = \{x_1, x_2, \cdots\}$. $A_x \subset Q$; Show: if $x \neq y$ then $A_x \cap A_y$ is finite. Now justify $|\Delta| = c$.

22. Consider the set $X = [0,1] \times [0,1] - \{(0,0); (1,1)\}$ (two corner points removed) with dictionary order. This means $(x,y) \leq (u,v)$ if $x < u$ or $(x = u$ and $y \leq v)$. How are the words arranged in a dictionary?
Show that X has no first point; no last point; between two different points there is some thing strictly in between; every non-empty subset which is bounded above has a supremum. Is this order isomorphic to R?
Suppose you considered all of $[0,1] \times [0,1]$ with dictionary order. Which of the above properties hold? Suppose you considered $R \times R$ with dictionary order. Which of the above properties hold? Suppose you consider $Z \times [0,1)$ with dictionary order. Which of the above properties hold?

23. Suppose (X,d) is a metric space. Let us define

$$d_1(x,y) = \min\{d(x,y), 1\}.$$

Show that d_1 is also a metric.
Show that $x_n \to x$ in d iff $x_n \to x$ in d_1.
Show that a sequence is Cauchy in d iff it is Cauchy in d_1.
Show that a set is open in the metric d iff it is open in the metric d_1.
Show that the metric d_1 is bounded, whether d is bounded or not.
Suppose that in taking the minimum above, if I took minimum with 0.0001 how do the answers to above questions change?

Start with a metric space (X,d). Instead of the above, define

$$d_2(x,y) = \frac{d(x,y)}{1 + d(x,y)}.$$

Test whether all the above statements are still valid for d_2.

24. Consider the function $\tan x : (-\pi/2, +\pi/2) \to R$. It is strictly increasing, onto and one-one. Let $\tan^{-1}$ be the inverse map. Let us define

$$d_1(x,y) = |\tan^{-1} x - \tan^{-1} y|; \qquad x, y \in R.$$

Show that d_1 is a metric. Show that it is equivalent to d, that is, $x_n \to x$ in d iff $x_n \to x$ in d_1.

Do you think the following is correct: A sequence is cauchy in d iff it is Cauchy in d_1.

25. Optional Challenge: not part of course

Let $X = Q$, the set of rational numbers.

We define, for $x \in Q, x \neq 0$, order of x as follows: If x is an integer, then $O(x)$ is the largest power of 7 that divides x. If $x = a/b$ rational (with a, b integers), then $O(x) = O(a) - O(b)$. Show that this definition does not depend on how you represent the rational as a fraction.

For $x \in Q$, we define $||x||$ as follows:

$$||0|| = 0; \qquad ||x|| = 7^{-O(x)} \quad x \neq 0$$

note the negative sign for exponent.

Show $(i)||x|| = 0$ iff $x = 0$.

$(ii)||x + y|| \leq ||x|| + ||y||$. $(iii)||xy|| = ||x||\ ||y||$.

Define $d(x, y) = ||x - y||$. Show that this is a distance on Q.

Let $x \geq 1, y \geq 1$ be integers. Show $d(x, y) \leq 1/7^n$ iff $x = y \bmod(7^n)$.

Calculate for $n, m \in Z$,

(i) $d(7^n, 7^m)$ (ii) $d(7^{-n}, 7^{-m})$

Wonder: $7, 7^2, 7^3, 7^4, \cdots \to 0$.

Convince yourself that this metric takes (apart from zero) only the values

$$\cdots, \cdots \frac{1}{7^3};\ \frac{1}{7^2};\ \frac{1}{7};\ 1;\ 7;\ 7^2,\ 7^3, \cdots\cdots .$$

Try to think which fellows are sitting exactly at a distance 1/7 from zero. Who are sitting at a distance 7 from zero.

A can of Fanta sold in London has 63 calories. Buy the same brand in India, and it contains three times as much sugar. It also includes artificial dye, whose presence in food products in Europe requires a prominent health warning. In India, the colorant's presence is noted merely in small print on the back of the can. The Food Safety and Standards Authority of India buckled after a tense March meeting with industry executives. It told the court in August that it was "difficult" to match international standards on packaging, echoing the industry position. Indian standard KitKats have 4.5% cocoa solids, while the milk chocolate in the Australian version contains at least 22% cocoa. All variants of Nestle's Maggi instant noodles sold in India are made with palm oil, whereas many versions sold in Britain use pricier sunflower oil. Revant Himatsingka made a clip that criticized the amount of sugar and artificial sweetener used in PepsiCo's Sting caffeine drink. PepsiCo successfully sought a take-down order of the video from a Delhi court. Economic Times; Aug 25, 2026

26. Here are some more examples of useful metrics on some useful spaces.

    (a) Consider a (finite) set S and an integer $N \geq 1$. Consider $X = S^N$, sequences (called strings) of length N from the set S. denote $x = (x_1, \cdots, x_N)$ for points in X. Put $d(x, y) = |\{i : x_i \neq y_i\}|$, that is, the number of places where the two strings differ. Verify this is a metric.

    This distance tells you how many places need to be changed to transform one sequence to the other. This is called Hamming distance and is useful in coding theory and DNA analysis. A large distance provides better error detection/correction in communication. A large distance between DNA strings of two animals will hint that they are different species.

    (b) Let $X = R^N$ and Σ be a symmetric positive definite matrix of size $N \times N$. Define $d(x, y) = \sqrt{(x-y)^t \, \Sigma^{-1} \, (x-y)}$. show that this is a metric on R^N. This is sometimes called Mahalanobis distance and is useful in Statistics. The points x and y are observations (may be vector consisting of nose length, cheek bone size, forehead width; scalp measurement etc of some skull). We have two observations on two skulls we found. Want to know if they belong to the same tribe. Here Σ comes from an assumed probabilistic model; appearance of its inverse seems mysterious but natural.

(c) Let X be the set of permutations of $S = \{1, 2, 3, \cdots, N\}$. For two permutations π and η define

$$d(\pi, \eta) = \left| \left\{ (i, j) : i < j; \begin{array}{lll} \pi(i) < \pi(j) & \& & \eta(i) > \eta(j) \\ \pi(i) > \pi(j) & \& & \eta(i) < \eta(j) \end{array} \text{ OR } \right\} \right|$$

that is; the number of pairs which are compared by π and η differently. Show that this is a distance. This is called Kendall's tau and is useful in statistics. Its largest value is $N(N-1)/2$. When is this value achieved? Suppose you and I rank (no ties) fifty selected hotels in Chennai. How different are our rankings? Above distance is one such measure.

(d) Let $X = \{(p_1, \cdots, p_N) : p_i \geq 0 \quad \forall\ i;\ \sum p_i = 1\}$. that is, all probability vectors. equivalently, all possible probability models for an experiment which has N outcomes. Define

$$d^2(p, q) = \frac{1}{2} \sum_1^N (\sqrt{p_i} - \sqrt{q_i})^2 = 1 - \sum \sqrt{p_i q_i}.$$

Then d is a distance, called Hellinger or Bhattacharya distance. This is useful in functional analysis (Hellinger) and in Statistics (Bhattacharya). How different are two models? Above is a measure.

(e) consider for a fixed $p > 1$, the space l_p of all complex sequences $z = (z_n : n \geq 1)$ such that $\sum |z_n|^p < \infty$. Show that this is a linear space and $d_p(z, w) = [\sum |z_n - w_n|^p]^{1/p}$ is a metric. You can consider the space l_∞ also. It is the space of all bounded complex sequences. That is, the space of all sequences $z = (z_n : n \geq 1)$ with $\sup |z_n| < \infty$. Show that $d_\infty(z, w) = \sup_n |z_n - w_n|$. is a metric on this space.

(f) $X = R^\infty = \{x = (x_1, x_2, \cdots) : x_i \in R; \quad i = 1, 2, 3, \cdots .\}$ Show $d_1(x, y) = \sum [\min\{|x_i - y_i|, 1\}/2^i]$. is a metric. Show $x^n \to x$ iff coordinate-wise convergence holds; $x_i^n \to x_i$ for each $i \geq 1$

27. For non-empty A, its diameter is $\delta(A) = \sup\{d(x, y) : x, y \in A\}$. In a sense, measures how far are the farthest corners of the town A from each other. Calculate diameters of the sets below.

(a) $X = R$. A is the interval $[0, 1]$; or the Cantor set; or set of rational numbers in $(0, 1)$; or set of irrational numbers in $(0, 1)$; or the interval $(0, \infty)$.

(b) $X = R^2$. A is a line segment; or a triangle; or the open unit disc; or closed unit disc; or ellipse (with semi-major axis a and semi-minor axis b); or the unit square $[0,1] \times [0,1]$.

(c) $X = C[0,1]$. A is the set of all x whose graph lies in the unit square; or set of all x whose graph lies in the unit disc.

(d) $x = R$ with discrete metric. A is the ball of radius $1/2$ around 5; or ball of radius $3/4$ around 5; or ball of radius 2 around 5; or ball of radius 5 around 5.

(e) Show that $\delta(B(a,r)) \leq 2r$. Do you think equality always holds? Show diameter of a set A is same as the diameter of its closure.

28. A set $A \subset X$ is said to be bounded if its diameter is finite. Emptyset by convention, is bounded. But anyway, let us consider non-empty sets only. Sometimes the following definition is used. A set A is bounded relative to a point $x \in X$ if $\sup\{d(x,a) : a \in A\} < \infty$, that is, there is a number M such that $d(x,a) \leq M$ for all $a \in A$.

    If A is bounded relative to one point, show it is bounded relative to any other point. Show that this definition is same as saying that the diameter is finite. Show that any Cauchy sequence is bounded. Show that union of finitely many bounded sets is bounded.

29. Let f be any continuous function on $[0,1]$. Let f_1 be any function whose derivative is f. Let, in general, f_n be any function whose derivative is f_{n-1}. (Of course, given f_{n-1} you have freedom to choose your favourite constant in getting f_n; no more!)

    Show the following: if $(\exists n)\ (\forall x) f_n(x) = 0$ then $f \equiv 0$.

    Show the following: if $(\forall x)(\exists n) f_n(x) = 0$ then $f \equiv 0$.

30. Consider $C[0,1]$. A function is said to be everywhere oscillating if it is not monotone on any (non-degenerate) interval. Pause. Can you picture such a function? Very very difficult. If I did not put continuity, it is trivial to imagine!

    There are many many everywhere oscillating functions. More precisely, Complement of the set of everywhere oscillating functions, in $C[0,1]$, is countable union of small sets.

I came into the room, which was half dark, and presently spotted Lord Kelvin in the audience and realised that I was in trouble at the last part of my speech dealing with the age of the earth, where my views conflicted with him. To my relief, Kelvin fell asleep. But as I came to the important point, I saw the old bird sit up, open an eye and cock a baleful glance at me. Then a sudden Inspiration came, and I said: Lord Kelvin has limited the age of earth, provided no new source was discovered. That prophetic utterance refers to what we are considering tonight – radium. Behold! The old boy beamed upon me!! — Ernest Rutherford
(recalling his 1871 address to the British Association of Science.)

31. (Principle of uniform boundedness) Suppose that (X, d) is a complete metric space. Let F be a collection of real valued continuous functions on X. Assume that the collection is point-wise bounded. that is, given $x \in X$, the set $\{f(x) : f \in F\}$ is a bounded subset of R. Show that there is an (non-empty) open set U on which these functions are uniformly bounded. That is, $\{f(x) : x \in U; f \in F\}$ is a bounded subset of R.

32. Let (X, d) be a metric space. Check: $\overline{A \cup B} = \overline{A} \cup \overline{B}$. Is $(A \cup B)^o = A^o \cup B^o$? Check $(A \cap B)^o = A^o \cap B^0$. Is $\overline{A \cap B} = \overline{A} \cap \overline{B}$? Show $(\overline{A})^c = (A^c)^0$. and $(A^o)^c = \overline{A^c}$
Show $\partial A = \overline{A} \cap \overline{A^c}$. Show $\partial(A \cup B) \subset \partial A \cup \partial B$ and $\partial(A \cap B) \subset \partial A \cup \partial B$. Equality does not hold in general. Give examples. Do you think $\partial(A \cap B) \subset \partial A \cap \partial B$. Do you think $A \subset B$ implies $\partial A \subset \partial B$? What is ∂Q where Q is the set of rationals in R. What is boundary of Cantor set in R? What is boundary of complement of Cantor set in R? What is boundary of a disc in R^2? boundary of a circle in R^2 boundary of a rectangle in R^2? Do you think A and $\overline{A}$ and A^0 all have the same boundaries?

33. Consider $X = C[0, 1]$ with the metric (show these are metrics) $d(x, y) = \int_0^1 |x(t) - y(t)|dt$ or $\rho(x, y) = \left(\int_0^1 |x(t) - y(t)|^2 dt\right)^{1/2}$. Show the space is not complete? Do you think convergence in any of these metrics is same as convergence in the sup metric? If A is a non-empty subset which is open and closed in this space then it must be either $\emptyset$ or all of X. Show this.

34. Let (X, d) be a complete metric space.
(a). Show that if $S \subset X$, then (S, d) is complete iff S is closed.
(b). Let U be an open subset of X. Consider any of the following

$$\rho(x,y) = d(x,y) + \left| \frac{1}{d(x,U^c)} - \frac{1}{d(y,U^c)} \right|; \quad x, y \in U$$

$$\rho(x,y) = d(x,y) + \min\left\{ \frac{1}{2^9}; \left| \frac{1}{d(x,U^c)} - \frac{1}{d(y,U^c)} \right| \right\}; \quad x, y \in U$$

is a metric on U. show that (U, ρ) is complete, AND convergence (and hence open/closed sets) in (U, ρ) is same as that in (U, d).
(c). Let $A = \cap U_n$ where each U_n is open in X. Thus A is intersection of countably many open sets. Put $d^*(x,y) = \min\{1, d(x,y)\}$. Put

$$\rho(x,y) = d(x,y) + \sum_n \min\left\{ \frac{1}{2^n}, \left| \frac{1}{d^*(x,U_n^c)} - \frac{1}{d^*(y,U_n^c)} \right| \right\}; \quad x, y \in A.$$

Show (A, ρ) is a complete metric space; d-convergence is same as ρ-convergence on A.
(d). The set of irrational numbers or the set of transcendental numbers in R is a possible candidate for above A. Show this.
(e). Here is a fact we shall not prove: Suppose $A \subset X$ and ρ is a metric on A such that (i) ρ-convergence is same as d-convergence on A and (ii) (A, ρ) is complete metric space. Then A must indeed be countable intersection of open sets in X.

35. Here is an application of Baire (a Diophantine approximation).

Suppose that $\{t_n\}$ is a strictly increasing sequence of positive numbers increasing to ∞. This is given to us. Then there is a large set S of real numbers such that if you take $x \in S$, then the set

$$\{t_n x + m : n \geq 1; m \in Z\}$$

is dense in R. of course, we can not exactly specify for which numbers x the above set is dense. However, Kronecker tells that when $t_n = n$ then the above set is dense for every irrational number x.

36. Here is another application (may not be easy but try) of Baire, similar to what we did for integrals. Suppose that f is a C^∞ function on $[0, 1]$.
If $(\exists k)\ (\forall x)\ f^{(k)}(x) = 0$ then f is a polynomial. Show.
If $(\forall x)\ (\exists k)\ f^{(k)}(x) = 0$ then f is a polynomial. Show this (not easy). In particular, there is one k such that the k-th derivative vanishes identically.

37. For a set A, let $l(A)$ denote set of limit points. $N = \{1, 2, 3, \cdots .\}$
(i). $A = \{(i, 0, 0, 0) : i \in N\} \subset R^4$. Show $l(A) = \emptyset$.
(ii) $B = A \cup \{(i, \frac{1}{j}, 0, 0) : i, j \in N\} \subset R^4$ Then $l(B) \neq \emptyset$ but $l(l(B)) = \emptyset$.
(iii) Let $C = B \cup \{(i, \frac{1}{j}, \frac{1}{k}, 0) : i, j, k \in N\} \subset R^4$. Then $l(C) \neq \emptyset$ and $l(l(C)) \neq \emptyset$ but $l(l(l(C))) = \emptyset$.
(iv) Let $D = C \cup \{(i, \frac{1}{j}, \frac{1}{k}, \frac{1}{m}) : i, j, k, m \in N\} \subset R^4$. Guess and show.
(v) It is not difficult to get similar sets in R itself. Try.

38. Giving explicit examples of nowhere differentiable continuous functions is not easy (though almost all paths of a Brownian particle are so). First such example was by Karl Weierstrass. We did one from F. Riesz. Here is one from W. Rudin: Let $\varphi(x) = |x|$ for $-1 \leq x \leq 1$ and $\varphi(x + 2) = \varphi(x)$ for all $x \in R$. Show $|\varphi(t) - \varphi(s)| \leq |t - s|$.
Define $f(x) = \sum_{n=0}^{\infty} (3/4)^n \varphi(4^n x)$. Show f is continuous.
Fix any $x \in R$. Let $\delta_m = \pm(1/2)\ (1/4^m)$. Show you can choose the sign for δ_m so that there is no integer between $4^m x$ and $4^m(x + \delta_m)$. We fix this sign now. Having fixed m, set

$$\gamma_n = \frac{\varphi(4^n(x + \delta_m)) - \varphi(4^n x)}{\delta_m}.$$

Show that $\gamma_n = 0$ for $n > m$ while $|\gamma_n| \leq 4^n$ for $0 \leq n \leq m$. Show

$$\left| \frac{f(x + \delta_m) - f(x)}{\delta_m} \right| \geq 3^m - \sum_{0}^{m-1} 3^n \geq \frac{1}{2}(3^m + 1).$$

Conclude that f is not differentiable at x.

39. Optional challenge: not part of course.
Here is an interesting algebraic field that builds on ultrafilter: algebraists and analysts are uninterested in this. Logicians have refined this idea thoroughly with profound consequences.
Let $\mathcal{U}$ be a free ultrafilter on $N = \{1, 2, \cdots\}$(Exercise 18). Let R_0 be the set of all infinite sequences $s = (s_n)$ of real numbers. Here are some examples of sequences for you to see. $s_n \equiv \sqrt{555}$; $t_n = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{n!}$. $u_n = n$; $u_n^* = n^2$; $w_n = 1/n$; $w_n^* = 1/n^2$.
Let us define $s \sim t$ if $\{n : s_n = t_n\} \in \mathcal{U}$. Show this is an equivalence relation and R^* be the space of equivalence classes. We define addition and multiplication on R^*: Take x and y in R^*, so equivalence classes.

Take one sequence $(s_n) \in x$ and one $(t_n) \in y$ and define $x + y$ to be the equivalence class containing the sequence $(s_n + t_n)$. Show this is a

good definition. Define multiplication in a similar way. Show that we have a field.

Define order on R^* as follows. Say $x \leq y$ iff $\{n : s_n \leq t_n\} \in \mathcal{U}$ where $(s_n) \in x$ and $(t_n) \in y$. Show that this is a good(?) definition. Show that R^* is a loset with this order. Show we have an ordered field.
if $a \in R$ define $\varphi(a) \in R^*$ to be the equivalence class containing the constant sequence with each term equal to a. Show that this is an embedding of $(R, +, \cdot, \leq)$ in $(R^*, +, \cdot, \leq)$ Thus this ordered field R^* contains usual R. Thus when we say the number $\sqrt[7]{33} \in R^*$ we mean the equivalence class containing the corresponding constant sequence.

Using the notation of sequences given at the beginning, show $[u] > k$ for each $k \in N$. In some sense $[u]$ is an infinite integer. u is an integer because $\{n : u_n \in N\} = N \in \mathcal{U}$ For each $\epsilon > 0; \epsilon \in R$, show that $0 < [w] < \epsilon$. In some sense $[w]$ is an 'infinitesimal', strictly positive but smaller than all ϵ's we know. So also is $[w^*]$. Show the infinitesimal $[w^*]$ is strictly smaller than the infinitesimal $[w]$.
We shall not discuss more about this field.

Leibniz quite rapidly developed formal Analysis in the form in which we now know it. That is, in a form specially suited to teach analysis by people who do not understand it, to people who will never understand it.

V I Arnold

I have come to realize that when Newton won the war against Leibniz, it was not Just credit at stake; it was a way of thinking about science. Leibniz had a more broader and philosophical view and saw calculus as not a specific tool in itself, but as an example that should inspire other kinds of universal tools. Had history developed differently there would probably be a direct line from Leibniz's 'Calculemus' to modern computation. Our own timeline followed a more circuitous path: 17th century breakthroughs led to an 18th century golden age of symbol pushing which led to a 19th century backlash obsessed with axiomatization and rigour which led to 20th century work on formal systems and computability which led to the 21st century laptop. Stephen Wolfram.

In what follows (X, d) is a metric space.

40. Show that $|d(x, A) - d(y, A)| \leq d(x, y)$. Argue that $x \mapsto d(x, A)$ is a uniformly continuous function.
Show that x is in the closure of A iff $d(x, A) = 0$.
If A is compact, then show that there is a point $u \in A$ such that $d(x, A) = d(x, u)$. Do you think when this equality holds A should be compact?
If A, B are disjoint closed sets show $f(x) = \frac{d(x,A)}{d(x,A)+d(x,B)}$ is a continuous function which is zero on A and one on B.
Let $C_b(R)$ be the collection of all real valued bounded continuous functions on R with sup metric. Show it is not separable.
Let X be a non-compact metric space. Show $C_b(X)$ with sup metric is not separable.

41. If A is compact and U is open and $A \subset U$ then show that there is a open set V such that $A \subset V \subset \overline{V} \subset U$. If A and B are disjoint compact sets show that you can get U, V open sets such that $A \subset U$ and $B \subset V$ and $\overline{U} \cap \overline{V} = \emptyset$.

42. Show $d(A, B) = \inf\{d(a, B) : a \in A\} = \inf\{d(b, A) : b \in B\}$.
Show that $d(A, B) \neq 0$ implies that $A \cap B = \emptyset$. Do you think the converse is true? Show $d(A, B) = d(\overline{A}, \overline{B})$.

Do you think $d(A,C) \leq d(A,B) + d(B,C)$?
if A and B compact, show, $\exists u \in A$, $\exists v \in B$, $d(A,B) = d(u,v)$.

43. Denote diameter of a nonempty set A by $\delta(A)$. Show $\delta(A) = \delta(\overline{A})$.
A set is said to be bounded if its diameter is bounded. Show that any Cauchy sequence is bounded. Show that union of finitely many bounded sets is bounded. Do you think every closed bounded set is compact?

44. $l(A)$ denotes set of limit points of A. $N = \{1, 2, 3, \cdots .\}$
(i). $A = \{(i, 0, 0, 0) : i \in N\} \subset R^4$. Show $l(A) = \emptyset$.
(ii). $B = A \cup \{(i, \frac{1}{j}, 0, 0) : i, j \in N\} \subset R^4$. Show $l(B) \neq \emptyset$; $l(l(B)) = \emptyset$.
(iii) $C = B \cup \{(i, \frac{1}{j}, \frac{1}{k}, 0) : i, j, k \in N\} \subset R^4$. Show $l(C), l(l(C)) \neq \emptyset$, $l(l(l(C))) = \emptyset$.
(iv). $D = C \cup \{(i, \frac{1}{j}, \frac{1}{k}, \frac{1}{m}) : i, j, k, m \in N\} \subset R^4$. Guess and show.
(v). get such sets in R itself.

45. If every real valued continuous function on X is bounded, show X is compact. If every real bounded continuous function attains its supremum then show that every real bounded continuous function attains infimum too. Show that when this happens the space is compact.

46. X is a compact metric space and f_n is a sequence of real continuous functions decreasing to a continuous function f point wise. Show that the convergence is uniform. This is known as Dini's theorem.

47. Describe *all* closed additive subgroups of R. Describe all open additive subgroups of R.

48. Consider $X = [0, 1]$ and let 2^X be the collection of all non-empty compact subsets of X with Hausdorff distance. Suppose K and L are singletons $\{x\}$ and $\{y\}$ guess what should be their distance and verify. Suppose K is a doubleton and L is a singleton. Then what is their distance? What if both are doubletons? What is the distance between $[0, 1/3]$ and $[2/3, 1]$. Guess first and proceed. Show that d_H is actually a metric on 2^X. show 2^X is compact.

49. M_{20} be the space of 20×20 matrices identified as R^{400}. Show multiplication of matrices is continuous. Show: set of invertible matrices GL_{20} is an open subset of M_{20}. Show that matrix inverse is a continuous map of GL_{20} to itself. In other words, the group operations are friendly with convergence. Since this is an open subset of a complete space you can treat it as 'complete' space (?). is this connected/compact?

SL_{20} is the subgroup of GL_{20} consisting of A with $|det(A)| = 1$, that is, $det(A)$ equals ± 1. show that this is a closed subset of GL_{20}. Since this is a closed subset, you can treat it as a complete metric space. Is this connected/compact?
What if I considered SL_{20}^{+}, matrices with determinant one.
What if I considered set of symmetric matrices in $GL(20)$? is it connected? If I considered P^{+} the set of symmetric positive definite matrices, is it open or closed? Is it connected?
What if I considered normal matrices, that is, matrices with $AA^t = A^tA$. Compact/connected?
If I considered the space O_{20} of orthogonal matrices, that is, matrices with $AA^t = A^tA = I$, identity matrix. Compact/connected?

50. Let X be a compact metric and Y be Polish. Let $C[X, Y]$ or simply C be the space of all continuous functions on X with values in Y.
Show $f \in C$ implies range of f is a bounded subset of Y.
Show $f \in C$ implies it is uniformly continuous.
Define for $f, g \in C$; $d_C(f, g) = \sup\{d_Y(f(s), g(s)) : s \in X\}$. Show that this is a metric and C is a complete metric space.
Let $X = [0, 1]$ and $Y = C[0, 1]$. Identify $C[X, Y]$ with $C([0, 1] \times [0, 1])$. What does identify mean here? Confused?

51. Not part of course, but easy.
Earlier we proved that e is not rational. Here is how you prove π is irrational. (Proof by Ivan Niven)
Let if possible $\pi = a/b$ ratio of positive integers. Define, for a positive integer n, two functions $f(x) = x^n(a - bx)^n/n!$ and
$F(x) = f(x) - f^{(2)}(x) + f^{(4)}(x) - \cdots + (-1)^n f^{(2n)}(x)$.
Show $f(x) = f\left(\frac{a}{b} - x\right)$.
Show that the function $n!f$ has integer coefficients.
Show that f and its derivatives have integer values for $x = 0$ and $x = \pi = a/b$.
Show $\frac{d}{dx}\{F'(x)\sin x - F(x)\cos x\} = F''(x)\sin x + F(x)\sin x = f(x)\sin x$.

$$\int_0^{\pi} f(x)\sin x dx = F(\pi) + F(0) \quad \text{is an integer.}$$

$$0 < f(x)\sin x < \frac{\pi^n a^n}{n!}; \quad 0 < x < \pi.$$

Note that f depends on n. Argue that if you choose n large enough the above inequality leads to a contradiction.

Academic Freedom Index report:
India's Central government expanded its authority over higher education, influencing university governance, leadership, appointments, and academic priorities. University administrators took actions that stifled campus discourse, including dismissing faculty members, cancelling academic events, and enforcing new policies restricting student expression.
Both students and police committed acts of violence at protests or other campus gatherings. Academic freedom was 'severely restricted' in 2026.

Government continued to deploy the University Grants Commission (UGC) as a "primary vehicle for state influence" over higher education.
The UGC faced scrutiny in recent years for policies perceived to tighten government control over universities, including loosening academic qualification requirements for vice-chancellors and accelerating the casualisation of academic labour by removing the 10 per cent cap on non-permanent, contract faculty.
India's AFI score has fallen sharply in recent years, from 0.38 in 2022 to 0.14 in 2025, according to AFI data. Telegraph India Sept, 2026.

52. Let S be any bounded closed rectangle contained in R^2. Show that the set of polynomials in x, y is dense in $C(S)$. Generalize to R^k.

53. The algebra generated by $\{1, x^2\}$ is dense in $C[0,1]$. Is it dense in $C[-1,1]$?
Suppose I take an integer $k \geq 1$. For what I is the algebra generated by $\{1, x^k\}$ dense in $C(I)$? Here I is a closed bounded interval.
Show Rational linear span of $\{e^{nx}; n \geq 1000\}$ is dense in $C[0,1]$.
Show Rational linear span of $\{1, (\sin x)^k; k \geq 79\}$ is dense in $C[0,1]$.

54. If f is real continuous on $[0,1]$ and $\int_0^1 f(x)x^n dx = 0$ for integers $n \geq 0$ show $f(x) \equiv 0$.

55. Here is a non-trivial application of the Stone-Weierstrass theorem. Let (X,d) be a compact metric space. Goal: to show that $C(X)$ is Polish.
(i) Assume: there is a sequence of functions $\{f_1, f_2, \cdots\}$ in $C(X)$ which separate points. Put $f_0 \equiv 1$ and $D_0 = \{f_n : n \geq 0\}$. Let D_1 be the collection of finite products of functions in D_0. Let D be finite rational linear combinations of functions from D_1. Show D is countable and closed under multiplication.
(ii). Let $\overline{D}$ be closure of D in our space $C(X)$. show that $\overline{D} = C(X)$.

Deduce $C(X)$ is separable.
(iii) Let $p \in X$ and $\epsilon > 0$. Put $f(x) = d(x,p)$. $g(x) = \min\{f(x), \epsilon\}$ and $h(x) = g/\epsilon$. Show these are continuous functions. Deduce: Given any open ball $B(p,\epsilon)$ there is a continuous functions f with $f(p) = 1$ and $0 < f \leq 1$ on B and $f(B^c) = 0$.
(iv) Show there is a sequence of open balls such that every open set is a union of some of these balls. Show that there is a sequence of functions as required in the first step.

56. Let X be a metric space which is not compact. show that the space $C_b(X)$ of bounded real continuous functions with sup metric is complete but not separable.

57. If $\{f_n\}$ is equicontinuous family on a compact metric space and $f_n \to f$ pointwise, show $f_n \to f$ uniformly.

58. Sometimes equicontinuity is defined as follows: Let X be a compact metric space and $S \subset C(X)$. Then S is equicontinuous if given $x_0 \in X$ and given $\epsilon > 0$, there is a $\delta > 0$ such that $f \in S, d(x_0, y) < \delta \Rightarrow |f(x_0) - f(y)| \leq \epsilon$. Show this is same as our definition.

59. Let $K(x,y)$ be a continuous function on $[0,1] \times [0,1]$. Let us define a map on $C[0,1]$ by

$$Tx(s) = \int_0^1 K(s,t)x(t)dt.$$

Show T takes the space to itself, in other words, if x is continuous then so is Tx (recall Analysis 1, but prove now).
Show: if $\{x_n\}$ is uniformly bounded sequence in $C[0,1]$ then the sequence $\{Tx_n : n \geq 1\}$ is pre-compact.

60. Let $A \subset C[0,1]$ be a collection of differentiable functions and there is a number M such that for all $x \in A$ and for all $t \in [0,1]$ we have $|x(t)| + |x'(t)| \leq M$. Show that A is pre-compact.

61. consider R^2 and $F(t,x) = x^{2/3}$ and $(t_0, x_0) = (0,0)$. Then $x(t) \equiv 0$ and $x^*(t) = t^3/(27)$ both solve $x' = F(t, x(t))$. Uniqueness of Picard is not available with Peano.

62. Let C be a closed subset of R. Show that we can express

$$C = D \cup P; \quad D \text{ countable}; P \text{ closed}; \quad P = \emptyset \text{ or } l(P) = P$$

Recall that $l(P)$ is the set of limit points of P and so the last phrase means that every point of P is a limit point of P. [take $(a,b) \cap C$ if countable and union of all such ab a, b vary].
Let P be a closed non-empty set in which every point is a limit point. Then show P must have same power as c. [Get two disjoint closed balls B_0 and B_1. within each get two disjoint closed balls B_{00}, B_{01} and B_{10}, B_{11}. Stare at these.

63. Unit balls
Remember: unit ball in R^k is $\{x : d(x,0) \leq 1\}$. Prove now.
Analogously, unit ball in $C[0,1]$ is $\{x \in C[0,1] : d(x,0) \leq 1\}$. Show that this is not compact.
unit ball in l_2 is $\{x = (x_n) \in l_2 : d_2(x,0) \leq 1\}$. Show this is not compact. However show that $\{x = (x_n) : (\forall n)|x_n| \leq 1/n\}$ is compact.

64. compact subsets of R^∞ Show that a subset $K \subset R^\infty$ is pre compact iff for each n, there is number M_n such that
$x = (x_n) \in K \Rightarrow |x_n| \leq M_n \ \ \forall n$.

65. If a metric space is totally bounded then show that every sequence contains a subsequence which is Cauchy. Thus, duty of totally boundedness is to provide a Cauchy subsequence; duty of completeness is to make it converge; together give compactness.

66. Stone-Weierstrass, Complex version
Let X be a compact metric space and $C[X, \mathfrak{C}]$ be the space of complex valued continuous functions on X. Let A be a complex sub algebra of this which contains constant functions and separates points. Suppose moreover that $f \in A$ implies its conjugate $\overline{f} \in A$. Show that A is dense in $C[X, \mathfrak{C}]$. Remember if $f(x) = f_1(x) + if_2(x)$ where f_1 and f_2 are real valued then conjugate of f is defined by $\overline{f}(x) = f_1(x) - if_2(x)$. (Reduce the problem to real case. If $f \in A$ show both f_1, f_2 are in A. Consider all $f \in A$ which are real and stare).

Midsem Friday — Sept 25 — 2 pm

Are our rulers against India, its future, its people and its culture?
(i) They call youth as future of India and They use bullets (Bihar), pellets and tasers (Delhi) on future of India. (ii) They behave barbarically with Indian citizens — push them into forest of neighbouring country; blindfolded, in the night, along with their children, (iii) Anything could be pretext for their lackeys to lynch Indian citizens – JSR or cow or (im)moral policing or whatever. (iv) 'sarva siksha abhiyan' means closing nearly one lakh schools in the country. (v) 'Beti bachao' means they receive obscene messages from slaves of our rulers (vi) They use 'dimagi naxal' to mean they hate free and independent thinkers. (vii) their motto is 'one religion, one education, one election, one food, one dress, one person' in the country, (viii) A chief minister talks like a goonda: "us or them", "either they stay or we do" about his citizenry. (ix) 'peaceful Sarbojanin Durga Puja' in Bengal makes way for 'violent Seemabaddha Durga Puja' $\cdots$ $\cdots$ $\cdots$

set of continuity points is exactly the set of rationals. Given a countable dense set $D \subset R$, there is no function f whose set of continuity points is precisely D.
For each interval $[a, b]$ with a, b rational, let f be any homeomorphism of $[0, 1] \to [a, b]$ and $C_{a,b} \subset [a, b]$ be the image of cantor set under this map. Let $D = \cup\{C_{a,b} : a, b \text{ rational}\}$. Show that there is no function $f : R \to R$ whose set of continuity points is exactly R. is this C countable?

69. Show the following: Every non-empty open set in R^2 is union, in a unique way, of countably many non-empty open connected sets.

70. Show that the set P of polynomials with rational coefficients is dense in $C[0, 1]$. Show that this set P is countable.

    If you are fed up of polynomials, here is another set. For rational numbers $0 = t_0 < t_1 < t_2 < \cdots < t_k = 1$ and rational numbers $r_0, r_1, \cdots, r_k$ let x be the function whose graph consists of straight lines joining (t_i, r_i) and (t_{i+1}, r_{i+1}) for $i = 0, 1, \cdots, k-1$. Let D be the set of all these function as $k \geq 1$ varies over integers and t_i and r_i vary over rationals as stated. Show that D is dense. Show D is a countable set.

71. Let Q be any bounded closed set contained in R^2. Show that the set of polynomials in x, y is dense in $C(Q)$. Generalize to R^k.

72. Consider l_2. Let D be the set of all sequences $x = (x_n)$ which have only finitely many non-zero terms and those terms are rationals. Show that this set is dense in l_2. Show that D is countable. Show that the same set is dense in l_1 too.

73. Just as we have $B(x, r) = \{y : d(x, y) < r\}$ ball around a point we also have ball around a set. For a non-empty set $S \subset X$, let $B(S, r) = \{y : (\exists\ x \in S)\ \ d(x, y) < r\} = \bigcup_{x \in S} B(x, r)$.

    If K is a compact and U an open set such that $K \subset U$ show that there is an $r > 0$ such that $B(K, r) \subset U$.

74. Let K be a compact and C a closed set in (X, d). If $K \cap C = \emptyset$ then show $d(K, C) > 0$. Recall $d(A, B) = \inf\{d(x, y) : x \in A;\ y \in B\}$.
    Do you think the statement remains true if both sets are closed?

75. Let $K(x, y)$ be a continuous function on $[0, 1] \times [0, 1]$. Let us define a map on $C[0, 1]$ by $Tx(s) = \int_0^1 K(s, t)x(t)dt$. Show that if x is continuous

then so is Tx. Show: if $\{x_n\}$ is a sequence in $C[0,1]$ which is uniformly bounded, then the sequence $\{Tx_n : n \geq 1\}$ is precompact.

76. Suppose that $A \subset C[0,1]$ is a collection of differentiable functions. Suppose that there is a number M such that for all $x \in A$ and for all $t \in [0,1]$

$$|x(t)| + |x'(t)| \leq M$$

Show that A is precompact.

77. identifying points
There is an important concept of 'quotient' in metric space.
Here is an example: $X = [0,1]$ Let us define equivalence relation: (i) $x \sim x$ if $x \neq 0, 1$ and (ii) $0 \sim 1$. Let us consider the space of equivalence classes X^*. Define a map $q : X \to X^*$ by $q(x) = [x]$ equivalence class containing x. Thus $q(x) = x = [x]$ or $x \neq 0, 1$ and $q(0) = q(1) = \{0, 1\}$. Declare a set $V \subset X^*$ open iff $q^{-1}(V)$ is open in X. This is called quotient space and the collection of open sets so defined is called quotient topology.
define $T = \{(x, y) \in R^2 : x^2 + y^2 = 1\}$ with usual Euclidean distance. Show that X^* is homeomorphic to T, via the map $h(x) = (\cos 2\pi x, \sin 2\pi x)$. First note that this is well defined on X^* because $h(0) = h(1)$.
Thus loosely speaking (also rigorously, if you make definitions), $[0,1]$ with zero, one identified is same as the unit circle.
Let $X = R$ and equivalence relation is $x \sim y$ if $x - y$ is rational. Show this is an equivalence relation, what are the equivalence classes and what si the quotient top[ology?

78. Write proofs of all the statements concerning complex numbers: field, modulus, convergence of sequences and series, orthonormal bases, exponential function, De Moivre formula et cetera.

In the math education community the term productive struggle describes the state of actively wrestling with a problem, persistently trying out various strategies, being willing to take risks, being unafraid of mistakes, and progressing incrementally in understanding the underlying ideas. This wrestling produces a certain kind of 'endurance' which enables us to be comfortable with the struggle. This endurance produces an 'unflappable character' that benefits us in addressing life problems calming us with the knowledge that it is okay if we do not solve a problem right away. We appreciate not solving a problem is as important as solving it, that, as Simone Weil suggested, the effort to grasp truth is itself worthwhile for increasing our aptitude, even if it produces no visible fruit. In the struggle, we acquire competence to solve new problems and fortify an expectant hope that we will one day solve them. And when we struggle and at last succeed, we build self confidence. Francis Su.

79. (i) plot conjugate $\overline{z}$ taking z in various quadrants.
(ii) Geometrically plot the set $\{z : |z| = 5\}$. Plot the numbers i and $-i$ and $e^{i\pi}$ and $e^{i\pi/2}$ and $e^{i\pi/4}$ and $e^{i\pi/100}$.
(iii) Show every complex number z can be written as $z = re^{i\theta}$ where $r \geq 0$ and $0 \leq \theta < 2\pi$. Show if $z \neq 0$ then this representation is unique. What goes wrong if $z = 0$.
(iv)Show if $\sum |z_n|$ converges then $\sum z_n$ converges.
(v) Prove fundamental theorem of calculus: If $f : [a, b] \to \mathfrak{C}$ is continuous, then $F(t) = \int_a^t f(u)du$ is differentiable and $F'(t) = f(t)$. (vi) Show integral is linear: for complex valued continuous f, g on $[a, b]$;

$$\int_a^b [f(t) + g(t)]dt = \int_a^b f(t)dt + \int_a^b g(t)dt;$$

$$\int_a^b \alpha f(t)dt = \alpha \int_a^b f(t)dt; \quad \int_a^b \overline{f(t)}dt = \overline{\int_a^b f(t)dt}$$

remember here α could be complex. Please be serious and justify all your steps. Argue that these hold for Riemann integrable bounded functions on $[a, b]$ whether periodic or not; whether continuous or not.

80. $\sqrt{ab} = \sqrt{a}\sqrt{b}$; take $a = b = -1$ to get $1 = \sqrt{1} = i \cdot i = -1$. What is wrong?

81. Calculate for integers m, n;

$$\int_0^{2\pi} \sin mx \sin nx dx; \quad \int_0^{2\pi} \sin mx \cos nx dx; \quad \int_0^{2\pi} \cos mx \cos nx dx$$

82. If $f : [0,1] \to \mathfrak{C}$ is any continuous function, show that there exist a sequence period one continuous functions (f_n) such that

$$\int_0^1 |f_n - f|^2 \to 0; \qquad \int_0^1 |f_n - f| \to 0 \qquad (\clubsuit)$$

Do you think you can get such a sequence (f_n) so that convergence holds in sup metric, that is, $\sup |f_n(x) - f(x)| \to 0$?
To understand how far the demand ($\clubsuit$) can be fulfilled, show that even if f is indicator function of an interval, you can do it. Try first indicator of interval $[a,\ b]$ with $0 < a < b < 1$.
If you can achieve ($\clubsuit$) for f and g show that you can achieve for $f + g$ and $17f$.

83. Show that the function $f(x) = \frac{1}{2}$ for $0 < x < 1/2$ and $f(x) = -\frac{1}{2}$ for $1/2 < x < 1$ and $f(0) = f(1/2) = 0$ is Riemann integrable. Calculate Fourier coefficients and Fourier series.

84. Fix $0 < \epsilon < 1/4$. Put $f(x) = \frac{1}{2\epsilon}$ for $-\epsilon \leq x \leq +\epsilon$ and zero for $\epsilon < x < 1 - \epsilon$. Calculate its Fourier coefficients and Fourier series.

85. If $f(x) = \min_n |x - n|$,calculate Fourier coefficients and Fourier series.

86. If $f(x) = |\sin \pi x|$, calculate Fourier coefficients and Fourier series.

Yamuna region irrigation Research report by Dinabandhu Sahoo, U. Delhi: nickel levels 181 times the approved limit in potatoes – molybdenum 68 times the limit in red radish – Nickel levels 97 times the limit in spinach, 65 times in onions – chromium levels 22 times the limit in mustard, 12 times in spinach, and 11 times in white radish – Lead levels exceeded the limit by 10 times in potatoes, onions and 7 times in spinach.
South Asia Network on Dams, Rivers and People report: Maharashtra's Nashik, Ambad Industrial area, Ground water contamination has reached alarming levels, with laboratory tests detecting (50.5 mg per litre) cancer-linked hexavalent chromium at more than 1,000 times the permissible (0.05 mg per Litre) drinking water limit.
Our rulers are disgustingly dishonest and careless.

No new exercises, giving a pause.
Go through all the notes and problems so far discussed carefully, read, understand, work out, at least think about them. Our job is not to classify material into easy/difficult; abstract/concrete etc. Our job is to believe that all of us are capable of thinking and do it. If you feel something is not understandable, you should ask me/Abhiram/classmates and make sure you are clear.
As John Medina says: When you are trying to drive a piece of information into your brain's memory systems, make sure exactly what the information means. Do not try to memorize the information by rote and pray the meaning will somehow reveal itself.

(i) when millions were disenfranchised, court's response: 'do not worry, we restore your voting rights after the elections' – makes no sense. Was supreme court complicit in asking citizens (in the guise of logical discrepancies) uncivilized questions like, why my father produced five children, or why my father produced me when he was above fifty years age! (ii) thousands complained when taken off voter list in two constituencies, courts response: it is only by-election – 'only'? (iii) Someone who boasts of skull breaking and having support from rulers will get bail easily but not peaceful protesting youth who did not break anything – perversion of justice? (iv) After the Election commissioner says that Supreme court approved a form, the court denies but does not ask why EC lied – powerful can lie. (v) when Government lawyer asks the judges 'don't you have trust in PM appointing EC', the judges do not warn him of stupidity of this argument. (vi) one judge says: why are you afraid of Government surveillance unless you have something to hide – Hitler regime? It is a shame that a country that produced Honourable RadhaBinod Pal has degraded to this level. Only hope is that few judges still exist who understand meaning of truth, justice and citizen.

87. compact subsets of R^∞
The diagonal argument has great potential.
Remember we made R^∞ the space of sequences of real numbers into a metric space. Convergence is just coordinatewise convergence. Show that a subset $K \subset R^\infty$ is pre compact iff for each n, there is number M_n such that
$x = (x_n) \in K \Rightarrow |x_n| \leq M_n \ \ \forall n.$

88. Here is a generalization of the space $C(X)$ we considered. for metric spaces X and Y we denote $C(X, Y)$ to be the space of all continuous functions on X into Y.

Let $X = [0, 1]$ and Y is Cantor set, describe $C[X, Y]$.

Let $X = [0, 1]$ and Y the set consisting of the vertical lines at $x = 2$ and $x = 4$ in the plane. Describe $C(X, Y)$. Are you able to imagine it as two copies of $C[0, 1]$ sitting side by side. Can you clearly specify the identification map?

Let $X = [0, 1]$ and Y be the set of complex numbers. Describe $C(X, Y)$ are you able to imagine it as $C[0, 1] \times C[0, 1]$? Can you write down the

identification map?

Let $X = [0,1]$ and $Y = C[0,1]$. Identify $C(X,Y)$ with $C([0,1] \times [0,1])$. What does identify mean here?

89. Let X, Y be metric spaces and X is compact. C stands for $C(X,Y)$. Show $f \in C$ implies range of f is a bounded subset of Y.

    Show $f \in C$ implies it is uniformly continuous. This means, given $\epsilon > 0$, there is $\delta > 0$ such that $d_X(s,t) < \delta \Rightarrow d_Y(f(s), f(t)) < \epsilon$.

    Define for $f, g \in C$; $d(f,g) = \sup\{d_Y(f(s), g(s)) : s \in X\}$. Show that this is a metric on C. Show C is complete iff Y is complete.

90. You know that the function: one at rationals and zero at irrationals is not continuous. Do you think it is a pointwise limit of a sequence of continuous functions?
    Prove the following: If $\{f_n\}$ is a sequence of continuous functions which converges pointwise to a function f then the set of continuity points of f is Large: complement of a first category set.

91. Here is a game called Banach-Mazur game: To start with a set $A \subset R$ is given. Here is the game: I pick a closed bounded nondegenerate interval I_1, Then you pick a closed nondegenerate interval $I_2 \subset I_1$. Then I choose $I_3 \subset I_2$ and so on. If $\cap I_n$ has at least one point of A, I win; otherwise You win.
    Show that if A is a finite set you win; show a strategy. (though I have not told you what a strategy is, guess)
    if A is the set of rationals, you win; exhibit a strategy.
    If A is a set of first category, you win; describe a strategy.

92. Let $f \in C_{per}[0,1]$, define $g(t) = f(59t)$ Calculate $\widehat{g}$ in terms of $\widehat{f}$.

93. Consider the group (under multiplication) of non-zero complex numbers. Show that any compact subgroup of this is contained in T, unit circle (that is, complex numbers of modulus one).
    Describe all compact subgroups of T. Show that any infinite subgroup of T is dense in T.

94. Let $n > 1$. Show
$$\sum_{k=1}^{n} e^{2\pi ik/n} = 0$$

95. Instead of period one functions, suppose I started with period 2π functions. Thus you fix an interval of length 2π. what will be (e_n), orthonormal; system? What will be Plancherel?

(i) Apply Plancherel to the following function

$$f(x) = \begin{cases} \frac{x+2\pi}{2\pi} & \text{if } -\pi \leq x < 0 \\ \frac{x}{2\pi} & \text{if } 0 \leq x \leq \pi \end{cases}$$

to deduce

$$\sum \frac{1}{n^2} = \frac{\pi^2}{6} \qquad \sum \frac{1}{(2n-1)^2} = \frac{\pi^2}{8}$$

(ii) Apply Plancherel to

$$f(x) = \frac{x^2}{2\pi} - x; \qquad 0 \leq x \leq 2\pi$$

to show

$$\sum \frac{1}{n^4} = \frac{\pi^4}{90}$$

(iii) Apply Plancherel to

$$f(x) = \frac{x^3}{4\pi^2} - \frac{3}{2}\,\frac{x^2}{2\pi} + \frac{1}{2}\,x \qquad 0 \leq x \leq 2\pi$$

to argue

$$\sum \frac{1}{n^6} = \frac{\pi^6}{945}$$

[I have not searched literature to see if there is a general algorithm]

(♠) An X-ray diffraction pattern, in principle, contains all the information necessary to solve the crystal structure. Without the knowledge of phases the solution cannot be obtained by a straightforward single-step process and mathematical manipulation of the data is necessary to yield information about atomic positions or about phases. The Madras school of crystallography has specialized for many years in techniques involving the use of Fourier series with various types of coefficients and this book is devoted to a description of such methods. — A reviewer of the book 'Fourier methods in crystallography' authored by G N Ramachandran FRS and R Srinivasan (1970).
(♣) The 1985 Nobel Prize in Chemistry was jointly awarded to Herbert A. Hauptman and Jerome Karle "for their outstanding achievements in the development of direct methods for the determination of crystal structure" using Fourier Methods.
(★) The Nobel Prize in Physics 1915 was awarded jointly to Sir William Henry Bragg and William Lawrence Bragg "for their services in the analysis of crystal structure by means of X-ray". They too suggested Fourier Methods.

The problems below are from internet. I have not checked.

96. $f(x) = 1$ if $-\pi < x < 0$ and $f(x) = 0$ if $0 < x < \pi$. show that FS in $(-\pi, +\pi)$ is

$$\frac{1}{2} - \frac{2}{\pi}\left[\sin x + \frac{1}{3}\sin 3x + \frac{1}{5}\sin 5x + \cdots\right]$$

Argue

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots$$

97. $f(x) = -1$ for $-\pi < x < 0$ and $f(x) = 1$ for $0 < x < \pi$. show that FS is given by

$$\frac{4}{\pi}\left[\frac{\sin x}{1} + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \cdots\right]$$

98. $f(x) = 0$ for $-\pi < x < 0$ and $f(x) = x$ for $0 < x < \pi$. FS on $(-\pi, +\pi)$ is given by

$$\frac{\pi}{4} - \frac{2}{\pi}\left[\cos x + \frac{1}{3^2}\cos 3x + \frac{1}{5^2}\cos 5x + \cdots\right]$$

$$+\left[\sin x - \frac{1}{2}\sin 2x + \frac{1}{3}\sin 3x - \cdots\right]$$

Argue

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots; \qquad \frac{\pi^2}{8} = 1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \cdots$$

99. $f(x) = x$ for $0 < x < \pi$ and $f(x) = \pi$ for $\pi < x < 2\pi$. FS on $(0, 2\pi)$ is given by

$$\frac{3\pi}{4} - \frac{2}{\pi}\left[\cos x + \frac{1}{3^2}\cos 3x + \frac{1}{5^2}\cos 5x + \cdots\right]$$

$$-\left[\sin x + \frac{1}{2}\sin 2x + \frac{1}{3}\sin 3x + \cdots\right]$$

Argue

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots; \qquad \frac{\pi^2}{8} = 1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \cdots$$

100. $f(x) = \pi - x$ for $0 < x < \pi$ and $f(x) = 0$ for $\pi < x < 2\pi$. show that FS on $(0, 2\pi)$ is given by

$$\frac{\pi}{4} + \frac{2}{\pi}\left[\cos x + \frac{1}{3^2}\cos 3x + \frac{1}{5^2}\cos 5x + \cdots\right]$$

$$+\left[\sin x + \frac{1}{2}\sin 2x + \frac{1}{3}\sin 3x + \frac{1}{4}\sin 4x + \cdots\right]$$

101. Fourier series of $f(x) = x$ on $[-\pi, +\pi]$. show that FS in $(-\pi, +\pi)$ is given by

$$2\left[\sin x - \frac{1}{2}\sin 2x + \frac{1}{3}\sin 3x - \cdots\right]$$

Deduce again

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots$$

102. $f(x) = x^2$ on $(-\pi, +\pi)$. show FS is given by

$$\frac{\pi^2}{3} - 4\left[\cos x - \frac{1}{2^2}\cos 2x + \frac{1}{3^2}\cos 3x - \cdots\right]$$

$$\pi^2/6 = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} + \cdots$$

103. $f(x) = -x(\pi - x)$ for $-\pi < x < 0$ and $f(x) = -x(\pi - x)$ for $0 < x < \pi$. Show that FS is given by

$$\frac{8}{\pi}\left[\frac{\sin x}{1^3} + \frac{\sin 3x}{3^3} + \frac{\sin 5x}{5^3} + \frac{\sin 7x}{7^3} + \cdots\right]$$

Argue

$$\frac{\pi^3}{32} = 1 - \frac{1}{3^3} + \frac{1}{5^3} - \frac{1}{7^3} + \frac{1}{9^3} + \cdots$$

Rulers that destroy books and book stalls are spherical idiots. The leftist ideology is much more important now than ever before. We spend crores and crores to build temples but surgeries and deliveries are still done under torchlight. We spend crores and crores to build taller and taller statues but women and children have to walk miles to fetch water. We spend crores and crores to build bullet train to Gujarat but tribals have to carry their sick on cot for miles to reach hospital. We spend crores and crores on one man's travels abroad but close nearly one lakh schools in the country. Replacing Mahatma Gandhi National Rural Employment Guarantee Act, it appears that our rulers snatched away, a right to employment, no matter how limited, that the working people in rural India had fought hard to earn for themselves.

104. Calculate $\langle \sigma_n, \sigma_n \rangle$, where σ_n is the Cesaro average of S_N appearing in Fejer's theorem.

105. If $F \subset C[0,1]$ is closed bounded and equicontinuous subset, show that there is a $g \in F$ such that $\int_0^1 g(x)dx \geq \int_0^1 f(x)dx$ for all $f \in F$.

106. is the set $\{f \in C[0,1] : f(x) > 0 \; \forall x \in [0,1]\}$ an open set in $C[0,1]$? is the set $\{f \in C_b(R) : f(x) > 0 \; \forall x \in R\}$ an open set in $C_b(R)$? (All sup metric)

107. A metric space is totally disconnected if given any $x \neq y$, there is a disconnection with $x$and y in different components – this means there is a set both open and closed which contains x but not y. Show that Cantor set is totally disconnected. (A subset of R always gets relative topology from R, unless contrary is stated).

108. We talked about X which is completion of (complex) $C[0,1]$ under the metric: $d(f,g) = \int_0^1 f\overline{g}$. Show: Indicator of every subinterval of $[0,1]$ is in X. Show: Indicator of every open set and indicator of every closed set is in X. Show: if f, g are in X so is $f+g$ and cf for any complex number c. Show that if $f \in X$ then so is its conjugate.

109. For $n \geq 1$, let $f_n(x) = x + \frac{1}{\sqrt{n}} \sin(nx)$; $-\pi \leq x \leq \pi$. And $f(x) = x$ for $-\pi \leq x \leq \pi$. Show $f_n \to f$ uniformly, but f_n' does not converge to f'.

110. Let $a > 0$ be a fixed number. Let f be any continuous function $f : [0, \infty) \to R$ satisfying $\lim_{x\to\infty} f(x) = 0$. Show that f can be approximated as closely as we like by a function of the form $h(x) = \sum_{n=1}^{N} c_n e^{-nax}$.

111. Prove that e^x is not a polynomial. (Saying that it is an infinite series will not do; $1 = \sum_{n\geq 1} 1/2^n$, one side infinite series and other side is not).

112. Show that the set $\{f \in C[0,1] : \sup |f(x)| \leq 1\}$ is not a compact subset.
Show that the family $\{\sin(nx) : n \geq 1\}$ is not an equicontinuous family of $C[0, \pi]$.

113. Let $\mathfrak{F}$ be the family of all polynomials of all degrees in one variable. Show that the family $\{P(x^3) : P \in \mathfrak{F}\}$ is dense in $C[-100, +89]$.
Consider the family $\{P(x^2) : P \in \mathfrak{F}\}$ What is its closure in $C[-1, 1]$.

114. Let $f \in C^\infty[0, 1]$.
Show that there is a sequence of polynomials $\{p_n\}$ such that $p_n \to f$ and $p_n' \to f'$ uniformly.
Show that the same is true if I wanted upto 10 derivatives.

All these problems are from Kenneth Davidson and Allan Donsig: Real Analysis with Real applications.